

Investor sentiment in gold and gold futures market: Evidence from ChatGPT-generated sentiment index

Yunchuan Sun^a, Ying Xu^a, Buxin Zhang^a, Betty Simkins^b

^a International Institute of Big Data in Finance, Business School, Beijing Normal University,
Beijing 100875, China

^b Department of Finance, Oklahoma State University, Oklahoma, USA

ABSTRACT

Using ChatGPT, we construct a daily investor sentiment index of the gold market based on big texts from social media and investigate how our measure captures the sentiment of the gold futures markets. The results indicate that positive gold sentiments significantly enhance gold futures returns and that the sentiment index can also predict short-term returns. Furthermore, the sentiment index matters more in the hedging function of gold futures during market crisis periods.

Keywords: ChatGPT; Gold sentiment; Gold futures.

Investor sentiment in gold and gold futures market: Evidence from ChatGPT-generated sentiment index

ABSTRACT

Using ChatGPT, we construct a daily investor sentiment index of the gold market based on big texts from social media and investigate how our measure captures the sentiment of the gold futures markets. The results indicate that positive gold sentiments significantly enhance gold futures returns and that the sentiment index can also predict short-term returns. Furthermore, the sentiment index matters more in the hedging function of gold futures during market crisis periods.

Keywords: ChatGPT; Gold sentiment; Gold futures.

1. Introduction

Recently, the investment landscape has witnessed a surge in interest and enthusiasm surrounding the gold futures market whose prices have been continuously rising (World Gold Council¹). At the time of writing this article on June 11, 2024, the international gold price based on LBMA AM had reached a peak of \$2444.35 per ounce on May 20, 2024, surpassing previous historical highs. The heightened attention can be attributed to a growing awareness of gold futures as a safe-haven asset to hedge economic uncertainties, financial crises, geopolitical tensions, etc. (Fang et al., 2018; Zhang et al., 2022). Due to factors such as arbitrage, supply and demand dynamics, and economic fundamentals, gold futures prices tend to converge with spot prices, and are influenced by investors' expectations of the spot gold market. The effect of investor sentiment has received much attention.

Sentiment metrics derived from texts, with their economic viability, effectiveness, and objectivity, are pivotal to understand investor behaviors and market trends. Smales (2014) finds that negative news sentiment has a more pronounced short-term impact on gold futures. In particular, achieving high *accuracy*, *precision*, and *recall*, ChatGPT has garnered significant attention due to its outstanding performance in text-based sentiment analysis (Julianto et al., 2023). ChatGPT is a powerful AI tool for analyzing texts, which can produce more consistent and accurate results than previous models. In this research, we develop gold sentiment metrics with ChatGPT and investigate the impact on gold futures.

The contributions of our research are as follows. First, with ChatGPT, we develop a novel daily investor sentiment index towards gold using texts from social media, which is a direct sentiment metric for investors. Second, the relationship between gold sentiment and the gold futures market is explored, shedding light on the influence of investor sentiment on market trends. Third, we show that gold sentiment can interpret gold futures performance, which contributes to an understanding of gold futures 'hedging function'. Also, we highlight the importance of sentiment analysis in actual investment and risk management strategies.

¹ <https://china.gold.org/>

2. Index Construction

2.1 Data collection

The Eastmoney *Guba* is a renowned stock community known for its large user base and rapid information updates in China, whose *HuangjinBa* (Gold Forum) is dedicated to the message sharing, discussion and analysis of the gold market. In order to capture investors' interest in gold, we crawled all post titles in *HuangjinBa* from 2010 to 2022, which include the number of views and comments, title, author, and last update time of each post. We analyze these to construct our investor sentiment gold index.

2.2 Index construction

2.2.1 Text sentiment analysis with ChatGPT

The release of ChatGPT is a significant breakthrough in the field of text analysis for artificial intelligence. By utilizing the Application Programming Interface (API), we conducted text analysis on investor posts in the *HuangjinBa* forum based on ChatGPT. Specifically, we prompted ChatGPT to quantify sentiment on a scale from -50 (extremely negative) to 50 (extremely positive).

We made the following conversations with GPT:

You are a gold futures investor and I will give you a post title. What emotions does this post title evoke in you? Please rate the mood on a scale of -50 to 50.

For positive emotions, rate them on a scale of 0 to 50, with the stronger the positive emotion, the closer the score is to 50. The less strong the positive emotion, the closer the score is to 0.

If it is a negative emotion, score it on a scale of -50 to 0, with the stronger the negative emotion, the closer the score is to -50. The less strong the negative emotion, the closer the score is to 0.

Take advantage of the 100-point spread to quantify the different levels of emotional words. Returns a number. The content returned must strictly conform to the format standards, and do not give additional content.

If no emotion can be deduced from the title, return 0.

A score of 0 indicates that the poster adopts a neutral position towards the gold market and demonstrates a lack of trading intention. Some posts are not related to gold futures, which will also be marked as 0 points. All posts scored 0 were removed from our full sample. Finally, we obtain a data set where each post is matched with its corresponding sentiment score.

2.2.2 Measurement of sentiment

Given the previous sentiment score for each headline, we next constructed the daily investor sentiment index. Referring to Sun et al. (2021), we adopt the following steps to do the measurement.

Firstly, we calculated the impact factor of each post. Posts with more readings and comments tend to have larger influence on other investors, so the influence factor of such posts is greater. We take one year as a period², and if there is a total of n_t posts during the period of t , where every post i has a number of readings denoted as $r_{i,t}$ and a number of comments denoted as

² There is little data in 2010, so we combined year 2010 and 2011 into one period to calculate.

$c_{i,t}$. So, we defined the influence factor $h_{i,t}$ of post i as follows.

$$h_{i,t} = \frac{\sum_{i=1}^{n_t} r_{i,t}}{\sum_{i=1}^{n_t} c_{i,t}} \times c_{i,t} + r_{i,t}$$

Next, we further smoothed the influence factor towards 1 and denote it as $H_{i,t}$.

$$H_{i,t} = \ln(h_{i,t}) - \frac{\sum_{i=1}^{n_t} \ln(h_{i,t})}{n_t} + 1$$

Thirdly, considering both the sentiment expression of each post and its external influence, we took the influence factor as a multiplier for the textual sentiment score from ChatGPT, and calculated the sentiment index $T_{i,t}$ of a single post.

$$T_{i,t} = H_{i,t} \times score$$

The data processing of posts on December 31, 2011 is shown in Table 1.

Table 1 Textual sentiment analysis for single post.

View	Comments	Title	Score	h	H	T
2874	2	“下周黄金预测 2” “Next Week’s Gold Forecast 2.”	0	3828.84	2.10	0
1433	0	“做空的人现在又担心死了” “The short sellers are worried sickly now.”	-20	1433.00	1.11	-22.27
1624	5	“这走势有点不讲道理了” “This trend is a little unreasonable.”	-30	4011.09	2.14	-64.28
2855	14	“1700 和 1800 敢做多的人到了 1530 附近居然要做空” “People who were long at 1700 and 1800 are going short around 1530.”	-30	9538.86	3.01	-90.27

Finally, we integrated all posts’ $T_{i,t}$ to get daily sentiment index. Assuming there are n_t posts in period t and n_s posts in day s , then the sentiment index $AUSenti$ of gold in day s is:

$$AUSenti = \frac{\ln(n_t)}{n_t} \times \sum_{i=1}^{n_t} T_{i,t}$$

In total, we obtained daily gold sentiment index including 4,459 entries of data.

2.3 Accuracy verification

After using ChatGPT to score, we randomly sampled data from 2015, 2018 and 2021 for manual scoring. We compared both results, and then trained ChatGPT to improve the accuracy, including artificially constructing dictionary, modifying prompt command, etc. The adjusted accuracy of the sample data can reach over 85%, which supports the validity of ChatGPT in sentiment rating.

3. Methodology

3.1 Data and variable description

We explore the relationship between gold sentiment and gold futures based on the daily time series data from 2013 to 2022. The dependent variable is the gold futures returns rate in the Chinese market (R), which is calculated by the daily closing price. The independent variable is our gold sentiment scored by ChatGPT ($AUSenti$). We also include control variables that may affect the price of gold spot and gold futures (Capie, 2005), including the spot gold price in the London market ($AULD$), the U.S. Dollar Index (DI), the New York Mercantile Exchange gold futures price ($AUCMX$), and the U.S. inflation rate ($USInflation$)³. In addition, we have excluded missing values, standardized data, and removed the 1% extreme observations. Table 2 reports the descriptive statistics for these variables.

Table 2 Descriptive statistics for variables.

<i>Variables</i>	<i>Obs.</i>	<i>SD</i>	<i>Mean</i>	<i>25p</i>	<i>Median</i>	<i>75p</i>
R	2080	0.01	0.00	-0.004	0.000	0.005
$AUSenti$	2080	0.21	0.03	-0.068	0.025	0.128
$AULD$	2080	260.30	1072	856.0	990.0	1324
DI	2080	6.25	94.94	92.61	95.69	97.94
$AUCMX$	2080	267.40	1441	1235	1313	1734
$US\ inflation$	2080	17.50	254.10	238.7	251.6	260.2

3.2 Baseline regression

To verify the impact of gold investor sentiment on the gold futures performance in the Chinese market, we use a baseline regression model:

$$R_t = \beta_0 + \beta_1 AUSenti_t + \beta_2 Controls_t + \varepsilon_t$$

where R represents the domestic gold futures yield, $AUSenti$ represents investor sentiment for gold, and its coefficient β_1 could reflect the extent to which the gold sentiment influences the domestic gold futures performance. $Controls$ represents the set of control variables. β_0 and ε_t are the constant and error terms, respectively.

³ Data Source: Choice's Global Macro database of non-quarterly CPI.

Table 3 Baseline regression results.

	(1)	(2)
	<i>R</i>	<i>R</i>
<i>AUSenti</i>	0.136*** (6.16)	0.143*** (6.40)
<i>AULD</i>		0.001 (0.03)
<i>DI</i>		0.019 (1.04)
<i>AUCMX</i>		0.034 (1.08)
<i>USInflation</i>		-0.027* (-1.74)
<i>constant</i>	0.377*** (27.49)	0.360*** (22.28)
<i>N</i>	2,080	2,080
<i>R – squared</i>	0.018	0.021

The values in brackets are t-values. ***, ** and * represent a confidence level of 1%, 5% and 10% respectively.

The variables have passed the *Johansen* cointegration test (p -value=0.001), and the regression results are shown in Table 3. According to the results, the regression coefficient of gold sentiment on domestic gold futures yield keeps significantly positive at the 1% level and the impact can be strengthened when controlling relevant factors. The numerical results indicate that average returns of gold futures will increase by 0.143% when gold sentiment increase by 1% and other factors remain unchanged. This suggests that positive investor gold sentiment can improve gold futures performance in the domestic market.

3.3 Predictive effect of lagged sentiment

In this section, we aim to explore the predictive effect of sentiment on the gold futures market, thus introducing lagged terms in the following model.

$$R_t = \alpha_0 + \alpha_1 \sum_{j=0}^{j=i} AUSenti_{t-j} + \alpha_2 Controls_t + \epsilon_t, i = 1, 2, 3 \dots$$

Table 4 presents the lagged regression results. According to columns (1), (3), and (5), it is found that lagged gold sentiment variables within two-period lag significantly predict gold futures return at the 1% level. With the incremental addition of lagged sentiment, the overall explanatory level of the model is also increasing (R^2 increases from 0.042 to 0.045, and then to 0.047). Moreover, the lagged one-period gold sentiment ($AUSenti_{t-1}$) has the highest regression coefficient, even compared with the contemporaneous sentiment index. The effect of lagged three-period gold sentiment ($AUSenti_{t-3}$) on the yield of domestic gold futures is no longer significant, suggesting that the positive influence is time-sensitive.

Table 4 Lagged regression results.

	(1)	(2)	(3)	(4)	(5)	(6)
	R	R	R	R	R	R
$AU\ Senti_t$	0.085*** (3.70)	0.092*** (4.01)	0.077*** (3.37)	0.085*** (3.71)	0.082*** (3.56)	0.089*** (3.85)
$AU\ Senti_{t-1}$	0.164*** (7.19)	0.171*** (7.46)	0.144*** (6.06)	0.150*** (6.30)	0.147*** (6.17)	0.152*** (6.39)
$AU\ Senti_{t-2}$			0.068*** (2.98)	0.075*** (3.27)	0.080*** (3.37)	0.085*** (3.58)
$AU\ Senti_{t-3}$					-0.043* (-1.86)	-0.037 (-1.61)
$AULD$		-0.004 (-0.12)		-0.006 (-0.19)		-0.005 (-0.16)
DI		0.020 (1.10)		0.020 (1.09)		0.020 (1.09)
$AUCMX$		0.043 (1.39)		0.047 (1.50)		0.045 (1.45)
$USInflation$		-0.026* (-1.69)		-0.025 (-1.64)		-0.026* (-1.66)
$constant$	0.307*** (18.42)	0.283*** (14.88)	0.282*** (14.97)	0.253*** (11.93)	0.297*** (14.51)	0.267*** (11.63)
N	2,079	2,079	2,078	2,078	2,077	2,077
$R - squared$	0.042	0.047	0.045	0.051	0.047	0.052

The values in brackets are t-values. ***, ** and * represent a confidence level of 1%, 5% and 10% respectively.

3.4 Analysis of Granger causality

The above analysis reveals that there is a correlated relationship between the sentiment level expressed in the posts of *Huangjinba* and the fluctuation of gold futures yields. Furthermore, we make Granger causality test to verify the causal relationship. The information criterion result shows that a four-period lag is better (AIC=-7.148*).

Table 5 Granger Causality test.

Y	X	df	$p-value$
R (futures yield)	$AUSenti$	4	0.000
	all	4	0.000
$AUSenti$ (sentiment)	R	4	0.112
	all	4	0.112

As shown in Table 5, the one-to-one causal analysis between gold sentiment and futures returns demonstrates the effectiveness of investor sentiment in explaining the performance of gold futures at the 1% level, while the lagged futures yield cannot effectively explain the

movement of *AUSenti*. That is, the investor sentiment for gold is the Granger cause of domestic gold futures yield, while the opposite is not plausible.

3.5 Hedging function of domestic gold futures

For most developed markets, the safe-haven effect of gold has been confirmed in studies. [Baur and Lucey \(2010\)](#) found that gold serves as a hedge against stocks on average and acts as a safe haven during extreme stock market conditions. [Hood and Malik \(2013\)](#) used sample data from the U.S. stock market and also supported gold's safe-haven effect. Especially during times of crises, economic shock often trigger panic among investors, drawing increased attention to gold. As a financial derivative, gold futures, with functions of price discovery and risk transfer, should serve as a significant complement to the hedging role of gold. To examine the hedging effectiveness of gold futures during market crises and delve deeper into the influence of investor sentiment, we selected four crisis periods: including the stock market crash in 2015, the US-China trade war in 2018, the COVID-19 pandemic in 2020, and Russia-Ukraine war in 2022⁴.

First, by DCC-GARCH model, we analyze the dynamic conditional correlation between daily gold futures yield and daily A-share returns in the Chinese market. Their correlation results, both within and outside the window periods, are shown in Table 6. Combining the results from panel A and B, we observe that during the crisis event period, the stock market performs poorly, while the gold futures show relatively strong performance. The negative correlation between gold futures and stock market volatility during the crisis periods significantly strengthens. Thus, we believe that when the market experiences negative exogenous shocks, gold futures could serve as a safe-haven asset to hedge against stock market risks.

Sun et al. (2021) have found that during the COVID-19 epidemic, there is a stronger positive correlation between individual investor sentiment and stock returns than usual. Following their discussion, we next examine the heterogeneous impact of investor sentiment on the gold futures market according to market crisis periods.

Table 7 reports the group regression results. It can be observed that gold sentiment keeps exhibiting a significant positive impact on gold futures returns. However, during market crisis periods, the coefficient of *AUSenti* increases ($p\text{-value}=0.0659$), indicating a stronger positive correlation between gold investor sentiment and domestic gold futures market returns. The results support that investor sentiment during crisis periods has a stronger explanatory power for the price movements in the gold futures market, even compared with the model with control variables included (R^2 increases from 0.013 to 0.033).

⁴ The four event windows are from June 15, 2015, to February 29, 2016, March 22, 2018, to December 13, 2019, January 23, 2020, to December 15, 2020, and February 15, 2022, to October 20, 2022, respectively.

Table 6 T-test for crisis event period results.

Panel A: T-test for crisis period returns.

	(A) Crisis = 0		(B) Crisis = 1		diff	t - value
	df.	Avg.	df.	Avg.	[mean(0) – mean(1)]	
<i>R – AUFuture</i>	1,162	0.0001	918	0.0005	-0.0004	-0.9773
<i>R – Ashare</i>	1,162	0.0017	918	-0.0003	0.0020	2.5172***

Panel B: T-test for DCC-GARCH model results.

Group	Obs.	Mean	Std. Dev.	Std. Err.
<i>(A) Crisis = 0</i>	1,161	-0.0057	0.0698	0.0020
<i>(B) Crisis = 1</i>	918	-0.0147	0.0817	0.0027
diff [mean(0) – mean(1)]		0.0090	t - value	2.6969***

Table 7 Group regression results.

	(1) Crisis = 0		(2) Crisis = 1	
	R	R	R	R
<i>AU Senti</i>	0.095*** (3.36)	0.101*** (3.52)	0.197*** (5.61)	0.212*** (5.95)
<i>AULD</i>		0.004 (0.11)		0.026 (0.48)
<i>DI</i>		0.009 (0.41)		0.068* (1.69)
<i>AUCMX</i>		0.040 (0.97)		0.022 (0.43)
<i>USInflation</i>		-0.031 (-1.09)		-0.061** (-2.38)
<i>constant</i>	0.400*** (22.49)	0.385*** (18.84)	0.343*** (15.92)	0.301*** (9.97)
<i>N</i>	1,161	1,161	918	918
<i>R – squared</i>	0.010	0.013	0.033	0.041

3.6 Robustness test

To test the robustness of the same-period effect of the investor gold sentiment on gold futures returns, we follow traditional metrics, measuring gold sentiment based on trading volume. We modify the independent variable by using the turnover rate (Turnover_t), trading volume (Volume_t), and position size (Pos_t) in the gold futures market as measurements for gold sentiment. The regression results are as follows.

As shown, the impact of the investor gold sentiment on the gold futures performance remains significantly positive. The results in the previous section are robust. However, the model's

fitness and coefficients decrease in column (1) and column (2), suggesting that the self-constructed textual sentiment data based on GPT in this study can provide a better explanation for gold futures returns.

Table 8 Robustness test.

	(1)	(2)	(3)
	<i>R</i>	<i>R</i>	<i>R</i>
Turnover	0.010* (1.66)		
Volume		0.014** (2.33)	
Pos			0.492*** (6.19)
<i>AULD</i>	0.002 (0.07)	0.002 (0.07)	0.006 (0.16)
<i>DI</i>	0.026 (1.26)	0.026 (1.26)	0.026 (1.29)
<i>AUCMX</i>	0.020 (0.58)	0.020 (0.58)	0.020 (0.58)
<i>USInflation</i>	-0.023 (-1.35)	-0.023 (-1.36)	-0.027 (-1.58)
<i>constant</i>	0.446*** (52.76)	0.446*** (52.79)	0.445*** (53.31)
<i>N</i>	1,591	1,591	1,591
<i>R – squared</i>	0.004	0.006	0.026

The values in brackets are t-values. ***, ** and * represent a confidence level of 1%, 5% and 10% respectively.

3.7 Addressing endogeneity issues

The occurrence of natural disasters such as earthquakes can cause people to panic, especially when earthquakes are frequent and produce serious secondary disasters. As a result, earthquakes can have a great impact on society and threaten people's lives, health and property safety, which in turn can impact their investment decisions, especially for the adjustment of portfolio risk levels. This paper argues that daily earthquake frequency nationwide⁵ can be used as an instrumental variable (IV) to solve endogeneity problems.

Specifically, we screen earthquake records with a magnitude greater than 3 and aggregate the daily count of earthquakes as our IV (*Earthquake*) and analyze using the two-stage least squares (2SLS) test. Table 9 displays the correlation matrix and regression results..

From the correlation matrix in panel A, we can preliminarily conclude that the *Earthquake* meets the criteria as an instrumental variable, and the F-statistic indicates that there is no issue of weak instrumental variables (p-value=0.000, minimum eigenvalue statistic=34.5286>10). Panel B reports estimation results of the first and second stages of 2SLS. Both models show

⁵ The data is sourced from www.harborn.cn.

significantly positive coefficients for the IV on the dependent variable, confirming consistency with previous findings.

Table 9 Estimation results of 2SLS.

Panel A: Correlation matrix.			
	<i>Earthquake</i>	<i>AUSenti</i>	<i>R</i>
<i>Earthquake</i>	1.000		
<i>AUSenti</i>	0.106*** (0.000)	1.000	
<i>R</i>	0.019 (0.234)	0.136*** (0.000)	1.000
Panel B: Estimation results of 2SLS.			
	<i>A: First-stage</i> <i>AUSenti</i>	<i>B: Second-stage</i> <i>R</i>	
<i>IV: Earthquake</i>	0.093*** (7.50)		
<i>AUSenti</i>		0.226* (1.74)	
<i>constant</i>	0.639*** (109.68)	0.307*** (3.62)	
Control Variables	YES	YES	
<i>Shea's adj. partial R2</i>	0.0388		
<i>Robust F</i>	47.8865 (0.000)		
<i>Minimum eigenvalue statistic</i>	34.5286		
<i>N</i>	2,080	2,080	
<i>R – squared</i>	0.041	0.019	

The values in brackets are t-values. ***, ** and * represent a confidence level of 1%, 5% and 10% respectively.

4. Conclusions

As one of the most prominent natural language processing models, ChatGPT holds great promise in academic research. In our study, based on social text data from the *HuangjinBa* forum in Eastmoney *Guba*, we employ ChatGPT's sentiment analysis technology to construct a daily investor sentiment index towards gold. The empirical findings are as follows. First, optimistic sentiment towards gold can significantly boost current gold futures yield in the Chinese market. Our robustness tests and endogeneity analysis (2SLS) also support these results. Second, investor sentiment for gold has the ability to predict gold futures returns in the short term and can be explained as a Granger cause for the performance of gold futures. Third, when the market is in a crisis period, gold futures can serve as a safe haven to hedge risks in the stock market, where investor sentiment plays a crucial role.

In conclusion, there is a close relationship between gold and the gold futures market, with

investor sentiment being a key factor. We show that ChatGPT can be used as a novel text analysis tool in financial research.

REFERENCES

- Baur, D. G., & Lucey, B. M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *The Financial Review* (Buffalo, N.Y.), 45(2), 217–229. <https://doi.org/10.1111/j.1540-6288.2010.00244.x>
- Capie, F., Mills, T. C., & Wood, G. (2005). Gold as a hedge against the dollar. *Journal of International Financial Markets, Institutions & Money*, 15(4), 343–352. <https://doi.org/10.1016/j.intfin.2004.07.002>.
- Fang, L., Chen, B., Yu, H., & Qian, Y. (2018). The importance of global economic policy uncertainty in predicting gold futures market volatility: A GARCH-MIDAS approach. *The Journal of Futures Markets*, 38(3), 413–422. <https://doi.org/10.1002/fut.21897>.
- Hood, M., & Malik, F. (2013). Is gold the best hedge and a safe haven under changing stock market volatility? *Review of Financial Economics*, 22(2), 47–52. <https://doi.org/10.1016/j.rfe.2013.03.001>.
- Julianto, I. T., Kurniadi, D., & Balilo, B. B. (2023). Enhancing sentiment analysis with chatbots: A comparative study of text pre-processing. *Journal Teknik Informatika* (Jutif).
- Smales, L. A. (2014). News sentiment in the gold futures market. *Journal of Banking & Finance*, 49, 275–286. <https://doi.org/10.1016/j.jbankfin.2014.09.006>.
- Sun, Y., Wu, M., Zeng, X., & Peng, Z. (2021). The impact of COVID-19 on the Chinese stock market: Sentimental or substantial? *Finance Research Letters*, 38, 101838. <https://doi.org/10.1016/j.frl.2020.101838>.
- Sun, Y., Zeng, X., Zhou, S., Zhao, H., Thomas P., & Hu, H. (2021). What investors say is what the market says: Measuring China's real investor sentiment. *Personal and Ubiquitous Computing*, 25, 587–599. <https://doi.org/10.1007/s00779-021-01542-3>.
- Zhang, X., Liu, J., & Zhang, G. (2022). Research on hedging effectiveness of gold futures at home and abroad in post-epidemic era. *Applied Economics and Policy Studies*. https://doi.org/10.1007/978-981-19-0564-3_59.